Advanced Seminar: Specialized Topics in Theoretical Physics

Bulk Topology in the Kitaev Chain

Schewtschik, Guilherme (3748052)

August 6, 2026

Abstract

This report summarizes the first block of the Advanced Seminar course quantum project. It discusses the foundations of the Kitaev chain and explains how bulk topology arises in this context. The report introduces the concept of Majorana fermions, discusses the Kitaev Hamiltonian, and presents the winding number as a topological invariant that distinguishes the trivial and topological phases of the system.

1 Introduction

First described by *Alexei Yu. Kitaev* in his 2000 paper [2], the Kitaev chain is a toy model used to study topological superconductors and topological quantum computing. It builds upon the work of the Italian physicist *Ettore Majorana*, whose concept of Majorana fermions gives rise to localized edge modes that provide a promising platform for topologically protected quantum computation.

The Kitaev chain exhibits a topological phase transition and serves as an ideal framework for understanding the relationship between superconductivity, Majorana edge states, and bulk topology. In particular, it demonstrates how a system's topological properties can be characterized by bulk invariants and how these invariants determine the existence of robust edge states through the bulk-boundary correspondence.

2 Majorana Fermions

Majorana fermions are a theoretical fermionic particle with the special property of being their own antiparticle, first proposed by *Ettore Majorana* in his 1937 paper *"A symmetric theory of electrons and positrons"* [3], where he describes chargeless spin$=\frac{1}{2}$ particles with real spinor fields.

In a field theoretical perspective we describe fermions with the so called fermionic annihilation and creation operators, denoted by $a(\mathbf{k}), a^\dagger(\mathbf{k})$ for particles and by $b(\mathbf{k}), b^\dagger(\mathbf{k})$ for anti-particles, such that their wave function is given by [4]:

$$\psi(x) = \sum_s \int \frac{d^3k}{(2\pi)^3 2\omega(\mathbf{k})} \left(a(\mathbf{k}) u_s e^{ikx} + b^\dagger(\mathbf{k}) v_s e^{-ikx} \right) \tag{2.1}$$

Since Majorana fermions are their own anti particle we can set $b(\mathbf{k}) = a(\mathbf{k})$, furthermore the charge conjugation operation imposes that $u_s^* = v_s$ [4], so we can rewrite the wave function.

$$\psi(x) = \sum_s \int \frac{d^3k}{(2\pi)^3 2\omega(\mathbf{k})} \left(a(\mathbf{k}) u_s e^{ikx} + a^\dagger(\mathbf{k}) u_s^* e^{-ikx} \right) \tag{2.2}$$

Rewriting the exponential terms as cosine and sine functions,

$$\psi(x) = \sum_s \int \frac{d^3k}{(2\pi)^3 2\omega(\mathbf{k})} \left((a(\mathbf{k}) u_s + a^\dagger(\mathbf{k}) u_s^*) \cos(kx) + i(a(\mathbf{k}) u_s - a^\dagger(\mathbf{k}) u_s^*) \sin(kx) \right) \tag{2.3}$$

So we define the operators $c_{A,s}$ and $c_{B,s}$ as follows:

$$c_{A,s} = a(\mathbf{k})u_s + a^\dagger(\mathbf{k})u_s^*, \quad c_{B,s} = i(a(\mathbf{k})u_s - a^\dagger(\mathbf{k})u_s^*) \tag{2.4}$$

This is the most general form of the Majorana operators, we can pin a representation by fixing a spinor basis $u_\uparrow$, $u_\downarrow$:

$$u_\uparrow = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_\downarrow = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

In this special case the form is independent of the spin index s, so we retrieve the representation used in Kitaev's original paper [2]:

$$c_A = a(\mathbf{k}) + a^\dagger(\mathbf{k}), \quad c_B = i(a(\mathbf{k}) - a^\dagger(\mathbf{k})) \tag{2.5}$$

One physical interpretation of the Majorana fermion is that the original fermion can be "split" into two Majorana fermions in the same physical position but at different "Majorana sites" [2], this is going to be of extreme importance when building the Kitaev chain in the next section.

It is also practical for the future to write our fermionic operators in terms of our Majorana operators:

$$a(\mathbf{k}) = \frac{c_A - ic_B}{2}, \quad a^\dagger(\mathbf{k}) = \frac{c_A + ic_B}{2} \tag{2.6}$$

3 Kitaev Hamiltonian

To construct our Hamiltonian we need to first define the physical system we are dealing with, in his original paper, Kitaev takes a chain of spin aligned electrons to be on top of a p-wave superconductor [2], this is crucial as the superconductor is what gives rise to the unpaired Majorana fermions in our system. By having the fermionic chain sit above of the p-wave superconductor this allows the fermions to pair in triplet like states, and by imposing an antisymmetric wave function, it causes the fermions to pair only with the adjacent sites in the chain. Furthermore the superconductor allows the electrons to be absorbed/released from the system in Cooper pairs such that only the parity of the system is preserved and not electron number.

With this taken into consideration Kitaev introduces our system's Hamiltonian [2]:

$$H = \sum_j -t(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j) - \mu(a_j^\dagger a_j - \frac{1}{2}) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \tag{3.1}$$

Here we annotate the fermionic creation and annihilation operators at the site j by a_j for $j \in [1, L]$ being the chain position index. The constants t, μ and Δ respectively correspond to the hopping term responsible for taking into account electrons moving between empty sites, the chemical potential term responsible for the addition and subtraction of fermion pairs from the system, and the induced superconducting gap pairing fermions at different spatial sites.

As Δ is in general a complex number $\Delta = |\Delta|e^{i\theta}$, we can incorporate this phase factor into our definition of the Majorana Fermions, with this we may express the Hamiltonian in the following way [2]:

$$H = \frac{i}{2}\sum_j -\mu c_{B,j}c_{A,j} + (t + |\Delta|)c_{B,j}c_{A,j+1} + (-t + |\Delta|)c_{B,j+1}c_{A,j} \tag{3.2}$$

We can see that the chemical potential term still pairs Majorana fermions at the same spatial site, while now the terms dependent on t and $|\Delta|$ now pair Majorana fermions at different spatial coordinates. This is crucial for the occurrence of free Majorana fermions at edge sites.

Taking the parameters t and $|\Delta|$ to be zero we arrive at the Trivial mode of the Kitaev Chain [2] where the only surviving term is the chemical potential and the Majorana fermions remain localized.

The Topological Mode occurs when $|\Delta| = t > 0, \mu = 0$, where our Hamiltonian becomes:

$$H = it \sum_j c_{B,j} c_{A,j+1} \tag{3.3}$$

Now the Majorana fermions pair with the adjacent spatial site, and we get unpaired fermions at $c_{A,1}$ and $c_{B,L}$, in Fig. 1 these are represented by red circles, while the paired fermions are represented by grey circles.

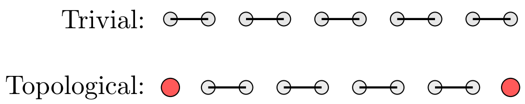

Figure 1: Majorana pairing patterns in the Kitaev chain.

However this is not restricted only to the case where $|\Delta| = t$, $\mu = 0$. In the thermodynamical limit we find that even for $\mu \neq 0$ there is still a topological phase given that $\Delta \neq 0$ and $|\mu| < 2t$ [1]. A deeper discussion of this will be held in section 4.

To continue the analysis of our Hamiltonian it is advantageous to take our fermionic operators into momentum space, and work in the reciprocal space from now on:

$$a_k = \frac{1}{\sqrt{L}} \sum_j a_j e^{ikj}, \quad a_k^\dagger = \frac{1}{\sqrt{L}} \sum_j a_j e^{-ikj} \tag{3.4}$$

With this our Hamiltonian becomes:

$$H = \sum_k (-\mu - 2t\cos(k)) a_k^\dagger a_k + i\Delta \sin(k)(a_k^\dagger a_{-k}^\dagger - a_{-k} a_k) \tag{3.5}$$

To find the spectrum of the Hamiltonian it is convinient to write it in Matrix form, one can do this by use of the Nambu vector $\mathbf{\Psi}_k$.

$$\mathbf{\Psi}_k = \begin{pmatrix} a_k \\ a_{-k}^\dagger \end{pmatrix}, \quad \mathbf{\Psi}_k^\dagger = \begin{pmatrix} a_k^\dagger & a_{-k} \end{pmatrix}$$

$$H = \sum_k \mathbf{\Psi}_k^\dagger h(k) \mathbf{\Psi}_k \tag{3.6}$$

Carring out the matrix multiplication we find what the Hamiltonian matrix $h(k)$ is:

$$h(k) = \begin{pmatrix} -\mu - 2t\cos(k) & 2i\Delta\sin(k) \\ -2i\Delta\sin(k) & --\mu - 2t\cos(k) \end{pmatrix}$$

Now findind the spectrum boils down to finding the eigenvalues of our matrix $h(k)$.

$$E_k = \sqrt{(\mu + 2t\cos(k))^2 + (2\Delta\sin(k))^2} \tag{3.7}$$

One is naturally interested in seeing how the energy differs between the two phases, and how the gap closes at the critical point. This is shown in Fig. 2 for different values of μ and t. With some simple algebra we can see that at $k = 0$ the energy reduces to $E_0 = |2t + \mu|$ and at $k = \pi$ we get $E_\pi = |2t - \mu|$, so the gap closes when $\mu = \pm 2t$.

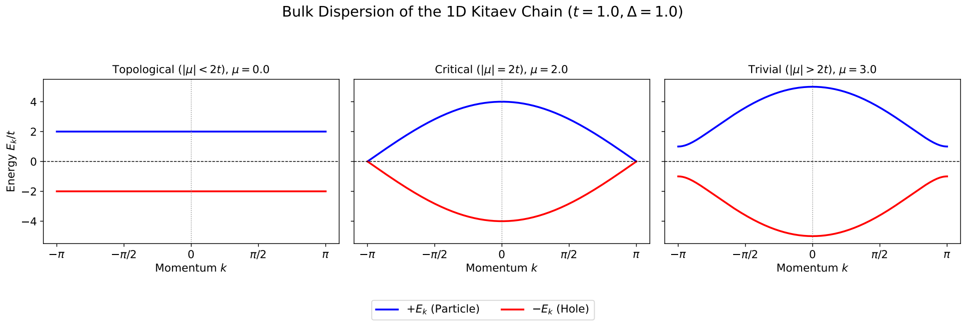

Figure 2: Bulk energy dispersion for different values of μ and t.

This gives us an idea of when and where the phase transition between the trivial and topological modes occurs. To further our understanding we first need a parameter that is solely dependent on the topology of the system, this is where the winding number comes into play.

4 Topology

For a given fixed energy E_k eq. 3.7 describes an ellipse. In the special case where $t = |\Delta|$ this becomes a circle centerd at $\frac{-\mu}{t}$, with radius $\frac{2E_k}{t}$. We can parametrize this circle with the vector $\mathbf{n}(k) = (2t\sin(k), \mu + 2t\cos(k))^T$ and see how it behaves for the different phases in Fig. 3.

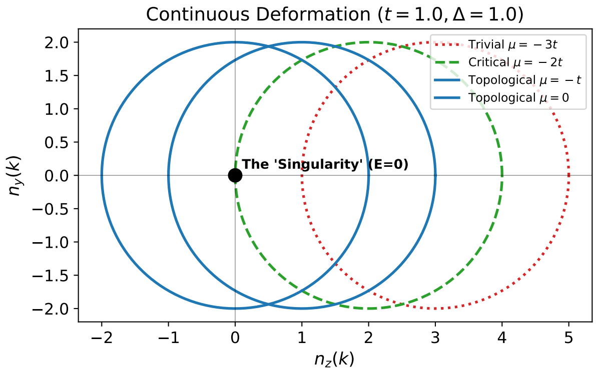

Figure 3: $\mathbf{n}(k)$ for different values of μ.

In the Topological phase *i.e.:* $|\mu| < 2t$ the vector $\mathbf{n}(k)$ encircles the origin, while in the Trivial phase *i.e.:* $|\mu| > 2t$ it does not. This indicates that we may gauge the different topological phases by determining wheter the curve $\mathbf{n}(k)$ encompasses the origin or not. To quantify this property we turn to the winding number ν, which is a topological invariant quantity that counts how many times a curve, in this case parametrized by $\mathbf{n}(k)$, winds around the origin.

$$\nu = \int \frac{d\theta_k}{dk}\frac{dk}{2\pi} \tag{4.1}$$

$$\theta_k = \arg((\mu + 2t\cos(k)) + i(2|\Delta|\sin(k))) \tag{4.2}$$

In Fig. 4 we can see that the winding number ν is 1 for the topological phase and 0 for the trivial phase, perfectly indicating the different phases.

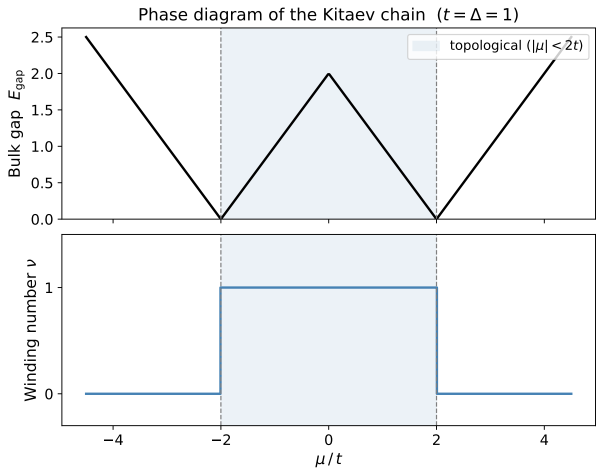

Figure 4: ν for different values of μ, alongside the Bulk energy gap.

5 Conclusion

The Kitaev chain models two distinct phases, the topological and trivial phases, giving us a way to study the transition between them. This occurs when the Bulk gap closes at the critical point $\mu = \pm 2t$. The topological phase is characterized by the presence of unpaired Majorana fermions at the edges of the chain. The different phases can be gauged by the winding number ν, a bulk topological invariant that distinguishes the two phases, giving us a straighforward way to determine the topological properties of the system.

Since the transition between the two phases requires the closing of the bulk gap, we can conclude that the topological phase is robust against small perturbations, as long as they do not close the gap. This robustness is why this toy model is used to study topological superconductors and topological quantum computing.

5.1 Large Language Models as a Tool for Physics Research

Throughout this course, one of our objectives was to evaluate and analyze how large language models (LLMs) can be integrated into physics research and the broader process of learning and scientific understanding.

In my experience, LLMs are a double-edged sword. On one hand, when used appropriately, they can significantly accelerate the learning process by gathering relevant resources, locating articles and reference material, and assisting with repetitive or time-consuming tasks. During this course, they proved particularly useful for compiling reports, preparing presentation slides, and supporting the implementation of code for the numerical simulations carried out in the later stages of the project.

On the other hand, excessive reliance on LLMs may lead to a false sense of understanding, as they can provide convincing explanations without encouraging the user to critically engage with the underlying concepts. They may also discourage the independent reasoning and analytical thinking that are essential when conducting scientific research. For this reason, I believe that LLMs should be regarded as tools that complement, rather than replace, traditional learning methods and critical thinking.

5.2 Acknowledgments

Throughout this report, the notation used for several variables differs from that adopted in the course slides, as I have predominantly followed the notation introduced by Kitaev in his original paper [2]. The use of large language models in the preparation of this report was limited to grammar correction, and assisting in locating relevant reference material.

References

[1] Loïc Herviou. "Topological Phases and Majorana Fermions". PhD thesis. Université Paris Saclay, 2017. URL: https://pastel.hal.science/tel-01651575/file/65288_HERVIOU_2017_archivage.pdf.

[2] Alexei Yu. Kitaev. "Unpaired Majorana fermions in quantum wires". In: *Physics-Uspekhi* (2000). URL: https://arxiv.org/pdf/cond-mat/0010440.

[3] Ettore Majorana. "Teoria simmetrica dell'elettrone e del positrone". In: *Il Nuovo Cimento* (1937). URL: https://inspirehep.net/files/ae2bc2372de056eb76b318ddbed3b61b.

[4] Mark Srednicki. *Quantum Field Theory*. Cambridge University Press, 2007. ISBN: 978-0-521-86449-7.